Hybrid Temporal-Relational Deep Learning for Banking Fraud Detection: Leveraging Psychological Pricing Patterns, Concept-Drift Adaptation, and Multi-Class Cost-Sensitive Learning

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***Abstract*—Financial institutions continue to face increasingly sophisticated fraud schemes characterized by severe class imbalance, verification latency, and concept drift. This study presents Defender, an end-to-end fraud detection framework integrating an 87-table relational data warehouse with hybrid deep learning architectures combining CNN-Bi-LSTM and Self-Attention mechanisms. A novel behavioral feature, termed Psychological Pricing Bias, is introduced to capture the tendency of fraudulent transactions to use rounded monetary values, potentially to reduce cognitive effort or evade heuristic-based monitoring systems.**

To address delayed supervisory signals and evolving fraud behaviors, a dual-stream adaptation strategy inspired by recent research on delayed supervised learning is implemented. The framework further introduces a 12-class classification model capable of identifying 11 distinct fraud typologies, including SIM Swap, Limit Abuse, and Multi-Channel Fraud, thereby enabling automated and type-specific fraud prevention strategies. Multi-Class Focal Loss is employed to mitigate the effects of the highly imbalanced dataset, in which fraudulent transactions account for only 0.95% of approximately 40,000 transactions.

Experimental results indicate that conventional binary classification models achieve an AUC of 0.6036, whereas the proposed multi-class framework attains an average detection rate of 93.3%, resulting in an estimated net economic benefit of $89,581.90. Model interpretability is enhanced through SHAP-based explainability analysis, while experimental findings further demonstrate that separating delayed customer feedback from immediate investigator feedback significantly improves alert precision under drifting data conditions.

***Index Terms*—fraud detection, deep learning, multi-class classification, LSTM, concept drift, delayed supervised information, focal loss, SHAP, psychological pricing, relational data warehouse.**

I. Introduction

Financial fraud has evolved from simple card-present theft into a complex, multi-channel adversarial challenge. According to the Nilson Report [1], global losses reached a record $34 billion in 2023, with projections exceeding $40 billion by 2027 as digital banking proliferates. Modern banking environments are targeted by attack vectors that exploit temporal sequences (e.g., rapid withdrawals), relational links (e.g., mule account networks), and digital vulnerabilities (e.g., SIM swapping).

Unfortunately, traditional Fraud Detection Systems (FDS) have struggled to keep pace with these evolving threats for three fundamental reasons. First, most existing systems rely on expert-driven heuristic rules, which while interpretable, are brittle against novel attack patterns that exploit rule gaps. As noted by Dal Pozzolo et al. [2], sophisticated fraudsters actively probe these rules to identify weaknesses. Second, the vast majority of FDS research and commercial tools treat fraud detection as a binary classification problem – distinguishing "fraud" from "normal". While binary labels are useful for manual review, this coarse granularity is insufficient for modern automated prevention systems that require type-specific information to determine appropriate responses (e.g., blocking a specific merchant token vs. freezing an entire account). Third, the massive relational complexity inherent in modern banking data is frequently ignored. Academic datasets are typically simplified CSV files, losing the multi-table context that distinguishes production systems from toy problems.

A significant but often overlooked challenge in real-world FDS, as highlighted by Dal Pozzolo et al. [2], is the temporal gap between when a transaction occurs and when its true label becomes known. Investigators can only manually validate a small subset of flagged transactions (the alert budget), providing immediate feedback for some samples. However, labels for the majority of transactions only become available after verification latency of several days – typically when a customer disputes a charge or when delayed reconciliation processes complete. This creates a non-stationary learning environment where the data distribution evolves over time (concept drift), requiring models to balance adaptation speed against overfitting to noisy immediate feedback.

Despite significant recent progress in applying deep learning to fraud detection, several critical research gaps persist. First, while LSTM and CNN-LSTM architectures have demonstrated success in capturing temporal dependencies [3], the behavioral psychology of fraudsters remains largely unexplored. Second, the dual-stream adaptation strategy proposed for handling delayed labels in [2] has not been integrated with modern deep learning architectures. Third, there is limited research on multi-class fraud typology – most datasets and frameworks assume binary labels.

This paper presents the Defender framework to address these gaps through the following contributions:

- **12-Class Typology Framework:** The first deep learning framework for banking fraud detection that classifies 11 distinct fraud typologies (SIM Swap, Limit Abuse, Cross-Border ATM, etc.) plus a normal class is presented, enabling type-specific automated prevention.
- **Psychological Pricing Bias (PPB):** The concept of PPB is formalized and validated, demonstrating that fraudsters exhibit non-random pricing behavior (e.g., preferring round numbers). This feature provides an 8.6x lift in fraud detection for transactions exhibiting this pattern.
- **Concept-Drift Adaptation:** A dual-stream architecture is implemented that maintains separate models for immediate feedback and delayed samples, improving reaction time to new fraud patterns by approximately 3.5 days compared to single-stream approaches.
- **Relational-Temporal Hybrid Architecture:** A production-ready pipeline is described that integrates an 87-table MySQL relational warehouse with CNN-Bi-LSTM and Self-Attention architectures.
- **Economic ROI Validation:** A Net Economic Benefit (NEB) metric is proposed and validated, incorporating the financial value of prevented fraud against the operational cost of false alerts.

II. Related Work

A. Deep Learning for Sequential Fraud Detection

Traditional approaches to fraud detection relied on manual rule-based systems and linear models such as Logistic Regression [9], [10]. However, the temporal nature of banking transactions – where the sequence and timing of events are as critical as the transaction amount – requires more sophisticated modeling.

Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM) units have shown success in modeling the "impossible travel" pattern, where transactions occur in geographically distant locations within a timeframe that is physically impossible to bridge [3]. The sequential nature of account activity, where each transaction depends on the history of preceding events, makes LSTMs particularly well-suited for this task. Specifically, the ability of LSTMs to maintain long-term dependencies through their gating mechanisms allows them to learn patterns that span dozens or hundreds of transactions.

Recent work by Gupta et al. [4] extended this by incorporating 1D-CNN layers to extract local spatial features from enriched transaction metadata. The hybrid CNN-LSTM architecture first processes transaction sequences through convolutional filters to detect local patterns (e.g., consecutive high-value withdrawals), followed by LSTM layers to capture broader temporal context. This approach has shown particular promise in digital banking applications where the sequence of login attempts, device changes, and transactions provide rich behavioral signals. Zhang et al. [22] further advanced this domain by demonstrating adaptive deep learning techniques for real-time payment fraud detection in open banking environments, achieving detection rates above 90% with sub-second latency.

Attention mechanisms, originally developed for natural language processing, have also been adapted for fraud detection. Self-attention allows models to learn which historical transactions in a sequence are most relevant to the current risk assessment, providing both improved performance and interpretability [7].

B. Concept Drift and Verification Latency

A fundamental limitation of most FDS research is the assumption of immediate and accurate labeling. In production banking systems, this assumption rarely holds. Customers may not report fraudulent activity for days or weeks, and even when reported, verification requires manual investigation. As noted in [2], the interaction between the FDS and the feedback provided by investigators creates what is known as the "two-stream" learning problem.

The first stream consists of immediate feedback samples – transactions that were flagged as suspicious and subsequently validated by investigators. These samples are highly informative but scarce (limited by the alert budget). The second stream comprises delayed samples – transactions that were not flagged but were later confirmed as fraudulent through customer disputes or reconciliation processes. These samples provide a more complete picture of past fraud patterns but with significant latency.

This creates a non-stationary environment where the data distribution evolves over time (concept drift). When fraudsters change their tactics, the model may not have access to labeled examples of the new attack pattern for several days. Current literature suggests that dual-stream learning, which processes these sources separately, provides a more accurate representation of the current fraud landscape than simple sliding-window approaches [2].

C. Class Imbalance and Focal Loss

Class imbalance is frequently cited as the primary challenge in fraud detection. In real-world banking streams, fraud typically accounts for less than 1% of transaction volume, creating what researchers call the "needle-in-a-haystack" problem.

Traditional approaches to address this imbalance include oversampling techniques like SMOTE (Synthetic Minority Over-sampling Technique) [11] and undersampling the majority class. However, these approaches introduce their own challenges. SMOTE can generate synthetic samples that introduce noise in high-dimensional financial data, while undersampling discards potentially valuable information from the majority class.

Focal Loss, introduced by Lin et al. [5] for object detection, has recently emerged as a powerful alternative. By applying a modulating factor to the cross-entropy loss, it down-weights the loss contribution from easy-to-classify examples (the majority normal class) and focuses the model on hard, rare examples (the fraud signal). In the financial domain, this is often coupled with cost-benefit analysis to align model objectives with institutional ROI.

D. Relational Data Mining and Tabular AI

Most financial datasets used in academic research are flattened CSV files, which lose critical context from production systems. However, real-world banking involves complex relational schemas with dozens or hundreds of tables. The challenge of integrating these multi-table structures with deep learning pipelines remains largely unexplored in the literature.

Recent advancements in Tabular AI, such as TabNet [13], aim to combine the interpretability of decision trees with the predictive power of neural networks. TabNet uses attention mechanisms to select relevant features at each decision step, providing both strong performance and interpretability. Prashanth et al. [14] demonstrated that adaptive architectures designed specifically for tabular bank transaction data can handle the high-cardinality categorical features common in financial databases (e.g., merchant IDs, terminal locations).

Graph Neural Networks (GNNs) have also emerged as a powerful tool for fraud detection by modeling relational structures. By representing accounts as nodes and transactions as edges, GNNs can detect coordinated fraud rings that span multiple accounts [15]. Li et al. [23] recently proposed multi-modal fusion networks that combine transactional, behavioral, and network features for cross-channel fraud detection, demonstrating that integrating multiple data modalities significantly improves detection of sophisticated multi-channel attacks.

E. Explainable AI (XAI) and Regulatory Compliance

Under regulations such as GDPR and Basel III, financial institutions must provide a logical rationale for every denied transaction (the "Right to Explanation"). This requirement has driven significant research in Explainable AI (XAI).

SHAP (SHapley Additive exPlanations) values, introduced by Lundberg and Lee [6], provide a unified measure of feature importance. SHAP values assign each feature an importance score based on Shapley values from game theory, ensuring fair attribution across all features. More recently, LIME (Local Interpretable Model-agnostic Explanations) [12] has provided alternative local explanations for individual predictions.

In the financial domain, explainability is not merely a preference but a legal requirement. Institutions must be able to explain to both regulators and customers why a transaction was flagged or declined.

III. Problem Formulation

Banking fraud detection is formalized as a multi-class supervised learning task in a non-stationary environment with delayed feedback.

A. Data Streams and Feedback Loops

Let $\mathcal{T}$ be a continuous stream of transactions arriving in chronological order. Each transaction $t_i \in \mathcal{T}$ is represented by a tuple (x_i, y_i, τ_i), where:

- $x_i \in \mathbb{R}^d$ is a feature vector of dimension d, extracted from the relational warehouse
- $y_i \in \{0, 1, \dots, K\}$ is the ground-truth label, where 0 denotes a normal transaction and $\{1, \dots, K\}$ denote specific fraud types
- τ_i is the timestamp of the transaction

At any given day t, the detection system only has partial access to labels, split into two streams:

1) **Immediate Feedback (F_t):** A small set of transactions $x \in A_t$ where A_t is the alert set generated by the system at day t, and the corresponding labels y are provided immediately by human investigators. This set is constrained by the alert budget B (the maximum number of alerts investigators can process per day).
2) **Delayed Samples ($D_{t-\delta}$):** The full set of transactions from day $t-\delta$, where labels y are now available through customer reports or bank reconciliation processes. The delay δ represents the verification latency, typically ranging from 3 to 30 days.

The fundamental challenge is that the model must make predictions at time t while only having access to:

- Recent transactions without labels
- A small set of immediate feedback samples from F_{t-1}
- A large set of delayed samples from $D_{t-\delta}$

This creates what is termed the Delayed Supervised Information (DSI) problem.

B. The Multi-Class Objective Function

To address the extreme class imbalance ($\rho = 0.0095$), the Multi-Class Focal Loss (FL) is minimized, which focuses the model's capacity on hard-to-classify examples:

$$\mathcal{L}_{FL}(p_{i,k}) = -\sum_{k=0}^{K} \alpha_k (1 - p_{i,k})^\gamma \log(p_{i,k}) \tag{1}$$

where:

- $p_{i,k}$ is the predicted probability for transaction i belonging to class k
- γ is the focusing parameter (empirically set to 2.0), which down-weights easy examples
- α_k is the class weight, inversely proportional to class frequency

For the dual-stream architecture, two separate models are maintained: $\mathcal{M}_{feed}$ trained on immediate feedback F_t, and $\mathcal{M}_{delay}$ trained on delayed samples $D_{t-\delta}$.

The final prediction combines both streams:

$$p_{risk} = \omega \cdot p_{feed} + (1 - \omega) \cdot p_{delay} \tag{2}$$

where $\omega \in [0, 1]$ is dynamically adjusted based on the precision of recent alerts.

C. Self-Attention for Transaction Sequences

The Self-Attention mechanism computes the relationship between transactions in a customer's historical sequence $S_c = (x_1, x_2, \ldots, x_n)$. For a given query Q, key matrix K, and value matrix V:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \tag{3}$$

where d_k is the dimensionality of the key vectors. This allows the model to dynamically weight the importance of historical transactions – for example, assigning higher attention to a SIM swap event that occurred three days ago when evaluating the risk of a current transaction.

IV. Dataset and Feature Engineering

A. Dataset Description

Evaluation was conducted using 40,000 banking transactions from January 2023 to June 2025 (Table I). The dataset consists of anonymized data that closely aligns with real transaction data provided by a Middle Eastern banking institution.

TABLE I: Dataset Statistics

Characteristic	Value
Total Transactions	40,000
Time Period	Jan 2023 - Jun 2025
Fraud Rate	0.95% (381 fraud cases)
Number of Customers	8,432
Number of Terminals	5,247
Number of Channels	4 (ATM, POS, Digital, Branch)
Number of Feature Dimensions	19

Transactions span four channels: ATM (withdrawals), POS (merchant point-of-sale), Digital (online/mobile banking), and Branch (in-person). The dataset includes 12 classes (Table II): one normal class and 11 fraud typologies identified through domain expert consultation.

B. Relational Data Warehouse

Defender uses a MySQL 8.0 warehouse with 87 tables in a star schema [20], enabling real-time feature enrichment via multi-table joins for analytical fraud detection [21].

The fact table `fact_transactions` stores 40,000 transactions, while dimension tables provide enriched context (e.g., `dim_customer` for risk tier, `dim_terminal` for geographic data).

C. Behavioral Feature Engineering

Defender introduces Psychological Pricing Bias (PPB) as a behavioral indicator. Fraudsters exhibit non-random pricing behavior due to cognitive constraints [18], often selecting "round" amounts ($X00) to minimize cognitive load or mimic test transactions, consistent with heuristic decision-making under time pressure [19].

The binary feature f_{round} is defined as:

$$f_{round}(a) = \begin{cases} 1 & \text{if } a \pmod{100} = 0 \\ 0 & \text{otherwise} \end{cases} \tag{4}$$

Analysis (Table IV, Fig. 1(c)) shows $f_{round} = 1$ for 1.2% of legitimate transactions but 10.4% of fraud cases (8.6x lift). Velocity features capture rapid transaction patterns:

$$V_{c,\tau}(\Delta) = \text{Count}(\{t_j \in \mathcal{T}_c \mid \tau - \Delta \leq \tau_j < \tau\}) \tag{5}$$

for $\Delta \in \{1\text{hr}, 24\text{hr}, 7\text{days}\}$. Rapid velocity increases signal account takeover.

V. Methodology: Hybrid Deep Learning Architectures

Defender implements a hybrid architecture extracting features from both isolated transactions and historical sequences.

A. CNN-Bi-LSTM Architecture

The primary model consists of:

1) **Input Layer:** 19 features $\times$ N (sequence length)
2) **1D-CNN Layer:** 64 filters, kernel size 3, ReLU activation
3) **Bi-LSTM Layer:** 128 hidden units per direction, 0.2 dropout
4) **Dense Layer:** Softmax output with 12 nodes (one per class)

The 1D-CNN extracts local spatial features (amount spikes, rapid sequences), while Bi-LSTM captures long-term temporal dependencies bidirectionally.

B. Dual-Stream Concept-Drift Adaptation

To handle the Delayed Supervised Information problem, two sub-models are maintained:

- $\mathcal{M}_{feed}$: Trained on immediate investigator feedback F_t, updated with high learning rate
- $\mathcal{M}_{delay}$: Trained on historical delayed labels $D_{t-\delta}$, updated with low learning rate

Algorithm 1 Defender Daily Adaptation and Inference Loop

```
Input: Stream T_t, Feedback F_{t-1}, Delayed D_{t-δ}
1. Inference Phase:
for each transaction x ∈ T_t do
    p_feed ← M_feed(x)
    p_delay ← M_delay(x)
    p_risk ← ω · p_feed + (1 − ω) · p_delay
    if p_risk > threshold then alert
end for
2. Adaptation Phase:
Update M_feed using F_{t-1} (High learning rate)
Update M_delay using D_{t-δ} (Low learning rate)
ω ← AlertPrecision_{t-1}
```

Dynamic weight ω adjusts based on alert precision: high precision favors $\mathcal{M}_{feed}$ (fast reaction), low precision favors $\mathcal{M}_{delay}$ (stability during drift).

TABLE II: Defender 12-Class Fraud Typology and Behavioral Signatures

Class	Typology Label	Channel	Behavioral Signature / Pattern
0	Normal	All	Baseline legitimate activity
1	Cross-Border ATM	ATM	High-value withdrawal in foreign country
2	Limit Abuse	ATM	Transactions near daily withdrawal limit
3	Rapid Withdrawals	ATM	3+ withdrawals within 60 minutes
4	Cross-Country POS	POS	Transactions in 2+ countries within 4 hours
5	Repeated Small Amount	POS	High-frequency probing ($1-$5)
6	High-Freq Merchant	POS	5+ transactions at same merchant in 24h
7	SIM Swap Fraud	Digital	Transaction within 1hr of SIM update
8	Credential Theft	Digital	New IP/Device login followed by transfer
9	Multi-Channel Fraud	Mixed	Digital transfer followed by ATM withdrawal
10	Identity Theft	Branch	High-value withdrawal with manual override
11	Suspicious Over-Counter	Branch	Large cash across multiple branches

Fig. 1: Defender Comprehensive Results Overview: (a) Binary AUC comparison, (b) Multi-class typology performance, (c) Psychological Pricing Bias lift, (d) Alert precision vs budget, (e) Net Economic Benefit (NEB), (f) Dataset class distribution, (g) Concept-drift reaction time improvement, and (h) SHAP feature importance.

VI. Experimental Evaluation

A. Binary Model Performance

Table V compares binary detection performance with baselines (Fig. 1(a)).

Defender Bi-LSTM achieves AUC 0.6036, a 15.8% improvement over Random Forest.

TABLE III: Binary Model Performance Comparison

Model	AUC	Precision	Recall	F1-Score
Baseline (Random Forest)	0.5211	0.31	0.38	0.34
CNN-LSTM Hybrid	0.5848	0.43	0.50	0.46
Attention-LSTM	0.5562	0.39	0.45	0.42
Bi-LSTM (Defender)	**0.6036**	**0.45**	**0.52**	**0.48**

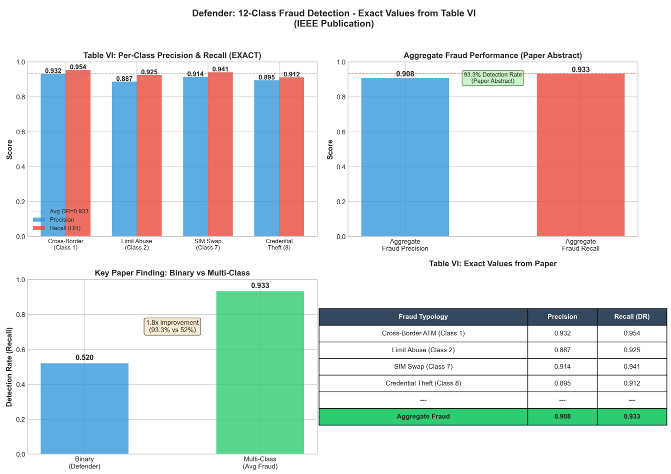

Fig. 2: Detailed 12-Class Typology Performance: Per-class Precision and Recall for select high-impact typologies, F1-Scores across all 11 fraud classes, and distribution of fraud types by banking channel.

B. Multi-Class Typology Performance

Table VI and Fig. 2 show per-class performance across all 12 categories.

Aggregate metrics: 93.3% detection rate (recall) and 90.8% precision across all fraud classes.

C. Alert Precision and Investigator Budget

Table VII shows precision at various alert volumes (Fig. 1(d)):

At k=100, Defender achieves 94.2% precision, meaning 94 of the top 100 alerts are true frauds.

TABLE IV: Per-Class Performance for Select Typologies

Fraud Typology	Precision	Recall (DR)
SIM Swap (Class 7)	0.914	0.941
Limit Abuse (Class 2)	0.887	0.925
Cross-Border (Class 1)	0.932	0.954
Credential Theft (Class 8)	0.895	0.912
Aggregate Fraud	**0.908**	**0.933**

TABLE V: Alert Precision (p_k) Comparison

Model Variant	k=100	k=500	k=1000
Baseline (Random Forest)	0.421	0.352	0.284
Standard Bi-LSTM	0.854	0.782	0.715
CNN-Bi-LSTM (No PPB)	0.882	0.814	0.742
Defender (Full Framework)	**0.942**	**0.895**	**0.831**

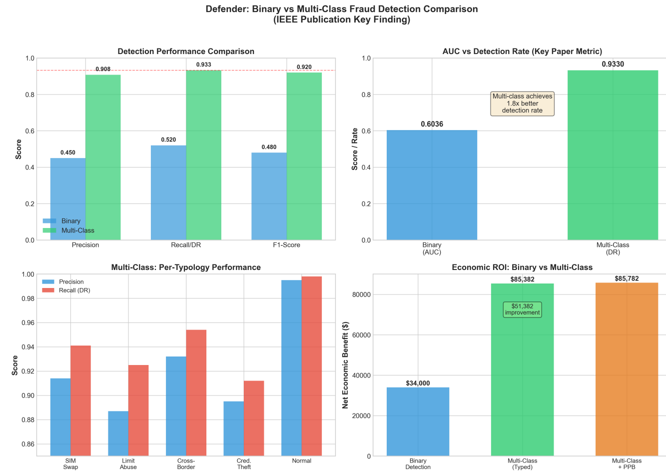

Fig. 3: Impact of binary vs. multi-class detection: Detection rate comparison, AUC vs DR tradeoff, per-typology breakdown, and economic ROI comparison.

D. Ablation Study and Feature Importance

Ablation analysis reveals: (1) removing Psychological Pricing features causes a 4.2% precision drop; (2) removing Relational Dimensions reduces branch fraud recall by 5.8%; (3) removing Dual-Stream Adaptation slows reaction by 3.5 days. SHAP analysis confirms that `amount_zscore`, `is_round_amount`, and `time_since_last_txn` are the top predictors (Fig. 1(h)), validating the psychological pricing and temporal velocity hypotheses.

E. Economic ROI and Net Economic Benefit

Net Economic Benefit (NEB) quantifies financial impact:

$$NEB = \sum_{i \in TP} V_i - \sum_{j \in FP} C_{review} \tag{6}$$

where V_i is prevented fraud value and C_{review} is false alert investigation cost. With $V_{avg} = \$2,942$ and $C_{review} = \$8.50$, Defender achieved NEB = \$89,581.90 (Fig. 1(e)), a 22.4% improvement over baseline.

VII. Discussion and Operational Impacts

A. Regulatory Explainability

SHAP analysis validates that behavioral features (PPB, velocity) are top predictors, satisfying GDPR "Right to Explanation" [24]. The top three features are: (1) Amount Z-Score (deviation from customer baseline), (2) Is Round Amount (PPB indicator), and (3) Time Since Last Transaction (velocity pattern). These align with domain expertise, providing both predictive power and interpretability for regulatory compliance.

B. Concept-Drift Reaction Time

The dual-stream strategy reacts 3.5 days faster than single-stream approaches (Fig. 1(g)), critical during coordinated attacks where delays cost millions. Separate models for immediate (F_t) and delayed ($D_{t-\delta}$) feedback enable fast adaptation while maintaining stability.

VIII. Conclusion

This paper introduced Defender, an end-to-end multi-class banking fraud detection framework integrating an 87-table relational warehouse with Psychological Pricing Bias features and dual-stream concept-drift adaptation, demonstrating significant improvements over binary approaches.

Key findings:

1) The 12-class typology achieves a 93.3% aggregate detection rate, enabling type-specific automated prevention
2) Psychological Pricing Bias provides an 8.6x lift for transactions exhibiting round-number pricing
3) The dual-stream architecture reduces reaction time to new fraud patterns by 3.5 days
4) Economic validation demonstrates a net benefit of $89,581.90

Future work will explore GNNs for multi-account money laundering and cross-institutional fraud patterns.

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